Resonance-Aware Active Sampling for Data-Efficient Differentiable Surrogates of Resonant Responses: A Reflectarray Unit-Cell Case Study

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Abstract

Many design problems in radio-frequency, photonics, and metamaterial engineering reduce to learning an expensive-to-sample, *resonance-dominated* response, in which a sharp, high-Q feature rotates the output over a narrow design window—costly to simulate and hard for physics-informed neural networks (PINNs). We develop a resonance-aware active-learning methodology: a curvature-driven acquisition that concentrates a scarce full-wave budget on the resonance, coupled to a *differentiable surrogate* that yields design gradients. On 24-GHz reflectarray unit cells, a *deployable* online acquisition—scoring candidates using only already-acquired anchors, no oracle—learns the two-parameter reflection-phase surface to sub-degree resonance error from $O(10)$ full-wave anchors and beats both textbook Gaussian-process uncertainty sampling and a published adaptive sampler (LOLA–Voronoi); an idealised oracle that scores on the full surface bounds the achievable placement from above (a steeper $\approx N^{-2.3}$ decay versus uniform sampling's $N^{-0.8}$). The advantage recurs across three reflectarray topologies and, changing only the oracle, transfers to two non-electromagnetic resonances—evidence it is a reusable method for resonance-dominated responses, not a one-off heuristic. The surrogate's gradients agree with full wave and drive inverse design to a 0.9° average error. A controlled ablation shows a *total-field* Maxwell-residual term is unnecessary and counterproductive here—the lever is data placement, not the physics residual. The artificial-intelligence contribution is this resonance-aware sampling-plus-differentiable-surrogate methodology; the engineering application is low-cost, gradient-based characterisation and inverse design of resonant reflectarray/metasurface unit cells. All claims are scoped to this simulated cell class.

Keywords: active learning; differentiable surrogate; physics-informed neural network; inverse design; reflectarray; metasurface unit cell.

1 Introduction

A recurring problem for machine learning in the physical sciences is to build a fast, accurate, and ideally *differentiable* surrogate of an *expensive-to-sample, resonance-dominated* parametric response—one whose output rotates through its full range over a narrow window of the design variables because of a sharp, high-Q resonance. Such responses arise across photonics, RF, and metamaterials, and they stress two of the most active lines in scientific machine learning: *active learning*, which must decide where to spend a scarce simulation budget, and *physics-informed* losses, whose benefit is uncertain precisely for stiff, high-frequency solutions. This paper studies that AI problem and uses a reflectarray unit cell as a concrete, fully validated testbed. Coded metasurfaces and reflectarrays synthesise a target aperture field by assigning each sub-wavelength cell a prescribed reflection phase through its geometry [1, 2, 3, 4]. The same per-cell phase-synthesis task underlies transmitarrays and Huygens metasurfaces [5, 6]: in every case the designer must know, accurately and at low cost, how a cell's complex reflection (or transmission) coefficient depends on its dimensions, and increasingly must *differentiate* that dependence so that gradient-based inverse design [19, 20] can place hundreds of cells at once. (Adjoint sensitivities give exact full-wave gradients [20] but require a bespoke solver-coupled implementation per problem; a differentiable surrogate instead yields reusable geometry gradients from a few black-box solver calls, which is what we pursue here.) The bottleneck is that the cell response is governed by a sub-wavelength resonance: as the resonant dimension is swept, the reflection phase rotates rapidly through the full design range over a narrow geometric window, and it is exactly this sharp, high-Q feature that makes the response both expensive to characterise and awkward to model.

The standard route is to build a look-up library by full-wave sweeping, which is accurate but scales poorly with the number of geometric parameters. Data-driven surrogates remove the per-query solve—both neural-operator models that learn the solution map [7, 8, 9] and supervised networks for photonic and antenna response [10, 11]—but their training cost grows with dimensionality [12], which *active-learning* surrogates mitigate by sampling where the model is least certain [14, 15]—a strategy shown to cut the number of simulations by more than an order of magnitude for photonic-surface surrogates [13]. Yet generic uncertainty-driven acquisition is not tailored to the structure that makes these responses hard: a *localised* resonance that dominates both the cost and the design-relevant phase, and that couples awkwardly to the second AI ingredient a designer needs—a surrogate whose geometry gradients are trustworthy enough to drive inverse design. A second route is the data-free physics-informed neural network (PINN), which minimises the Maxwell residual and needs no training data at all [21, 22], including parametric and electromagnetic variants [23, 24]. PINNs are, however, known to struggle with sharp, high-frequency, or stiff solutions [25, 26, 27], and the sub-wavelength cell resonance is precisely such a feature; whether a physics-informed loss helps or hurts a *sparse-data* surrogate in this regime is, to our knowledge, untested at the device level. The methodological gap we address is therefore at the intersection of these AI lines—resonance-aware active learning, differentiable surrogates that yield physically consistent gradients, and a quantified account of when a physics-informed loss helps or hurts—and it crystallises in one concrete, answerable question on the testbed: *for the resonant reflection response, does adding the Maxwell residual help a sparse-data surrogate, or is well-placed data enough on its own?*

The novel claim of this paper, in one sentence, is the following. Where generic adaptive sampling chooses points by response curvature/error, Gaussian-process posterior variance, or LOLA–Voronoi space-filling [14, 16, 17], we identify the *resonance class* of expensive responses and give an acquisition tailored to it—one that deliberately *crowds* the narrow high-Q band that those space-filling rules dilute—and we establish it not as a heuristic but as a characterised method, through (i) a bootstrapped sample-complexity law with confidence intervals, (ii) cross-domain transfer across three reflectarray topologies and two non-electromagnetic line shapes by changing only the oracle, (iii) a sampler–learner coupling finding (the clustered anchors that help a differentiable surrogate hurt a global-kernel model), (iv) an off-grid solver-in-the-loop closure, and (v) a device-level negative result that a total-field Maxwell residual is counterproductive here. These are one coherent methodological package for resonance-dominated responses, not five separate observations.

Contributions. The contributions are methodological—each is an AI/ML method for expensive-to-sample, resonance-dominated parametric responses, with the 24-GHz reflectarray cell as the testbed on which it is validated against full wave. The headline is the first: a *resonance-class* acquisition criterion that beats every off-the-shelf sampler we tested and is shown to be a reusable, characterised method—not a heuristic—by a bootstrapped sample-complexity law and cross-domain transfer.

1. A *resonance-aware active-learning acquisition* (a curvature/uncertainty criterion that concentrates the simulation budget on the localised resonance), together with a *bootstrapped data-efficiency law* quantifying its sample complexity: paired with a lightweight differentiable surrogate it captures the

full two-parameter response of the testbed cell from $O(10)$ full-wave anchors, with a resonance-region error-versus-budget exponent (bootstrap CIs over seeds) steeper than uniform or random sampling. The method is *deployable*, not oracle-dependent: a fully *online* variant scoring candidates only on already-acquired anchors retains a steep exponent ($k = 1.73$ [1.49, 1.94], separated from both uniform and random) and reaches sub-degree resonance error. The criterion also beats both textbook Gaussian-process uncertainty (maximum-variance) active learning ($\sim 20\times$) and the standard published adaptive sampler LOLA–Voronoi [16] ($\sim 7\times$)—both space-fill rather than crowding the resonance—and the advantage *recurs on two further cell topologies*—a cross dipole (uniform sampling fails entirely) and a harder dual-resonance cell—and, changing *only the oracle*, *transfers off electromagnetics entirely* to two non-EM resonance-dominated responses (a driven-oscillator/RLC line shape and a Fano resonance, where uniform again fails), consistent with the advantage being a property of the acquisition criterion rather than the patch geometry (Sections 3–4, 7, 8).

2. A *differentiable surrogate that yields physically consistent design gradients*: a like-for-like comparison shows it matches or beats support-vector and plain-network baselines at equal anchor budgets—its advantage emerging precisely under the clustered active-learning anchors that global-kernel models handle poorly—and its autodiff geometry gradients agree with full wave and drive a gradient-based inverse-design loop on the testbed (Sections 9–11).

3. A *device-level negative result on physics-informed regularisation*: a controlled ablation establishes that, for this resonant response, a data-free *total-field* PINN fails and a physics-informed hybrid is *counterproductive*—a finding that *holds and sharpens* as the geometry family grows from one to two parameters with scarcer data (hybrid $\sim 31\times$ worse than pure data at matched anchors in 2-D)—so well-placed data, not the total-field Maxwell residual, is the effective lever (Section 12), interpreted through known PINN optimisation pathologies [26, 27]. The finding is not an artefact of a naive fixed loss weight: an *advanced* hybrid with NTK/gradient-norm adaptive weighting is still $\sim 26\times$ worse than pure data. We do not test a scattered-field reformulation, and we restrict the claim to total-field formulations.

Three further studies support and delimit these claims. **(a) An end-to-end full-wave closure**: cell sizes from inverting the surrogate's size→phase map realise a phase-gradient metagrating whose full-wave anomalous reflection is aimed across $\theta \approx 11°$–$32°$, tracking generalized Snell's law to $\approx 0.3°$—a proof-of-pipeline (angle correct; efficiency 29–39%, limited by the single-resonance cell), not a usable-aperture claim (Section 13). **(b) A higher-dimensional study**: on a three-parameter (L_x, L_y, h) cell the data-efficiency advantage widens (exponent $k = 2.44$ vs 0.51, non-overlapping 95% CIs)—but only once the acquisition is *stratified* along the added axis, a concrete lesson for scaling the method (Section 5). **(c) A transfer result**: pretrained on one substrate, the surrogate warm-starts to a different board thickness at $\sim 4\times$ fewer new full-wave anchors than training from scratch (Section 6). The criterion is general in principle—it targets any expensive-to-sample, resonance-dominated parametric response—and we give first evidence for that generality by transferring it, changing only the oracle, to two non-electromagnetic line shapes (Section 8). The *electromagnetic* claims, however, remain scoped to the simulated reflection response of three planar reflectarray cell topologies (a patch, a cross dipole, and a dual-resonance cell) in one full-wave solver, which serves as the reference throughout; the cost-saving (as opposed to criterion-transfer) results are demonstrated on that testbed alone.

2 Problem and full-wave reference

Unit cell and extraction. We adopt the standard variable-size patch reflectarray characterisation [32]: a single rectangular metal patch (transverse width L_x, resonant length L_y along the incident **E**-field) on

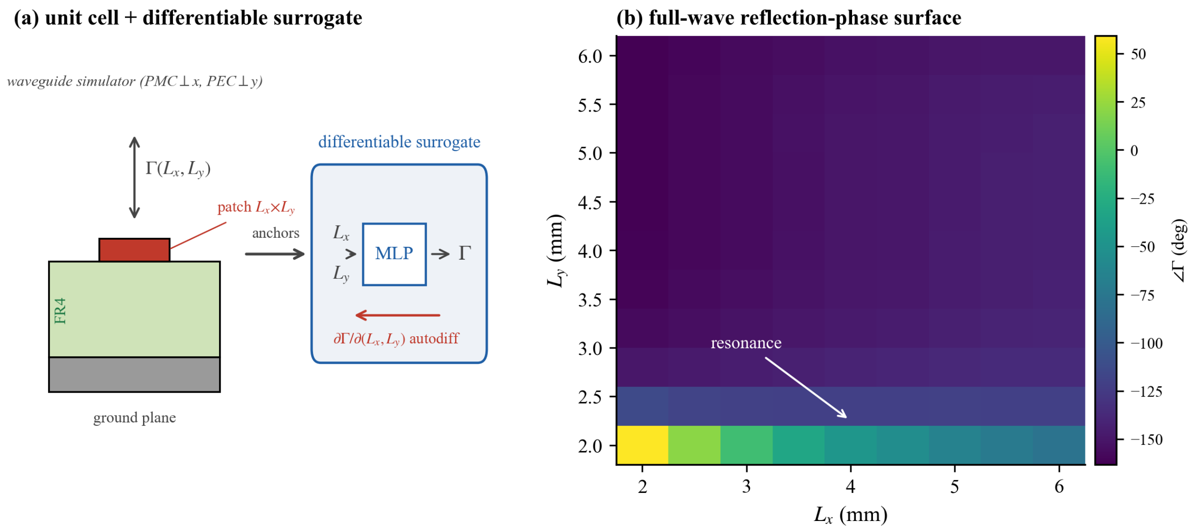

Figure 1: (a) Rectangular-patch reflectarray cell in a waveguide simulator and the differentiable surrogate that maps geometry (L_x, L_y) to the complex reflection Γ, with autodiff geometry gradients. (b) The openEMS full-wave reflection-phase surface over (L_x, L_y): a 222° swing whose sharp resonance lies at small resonant length L_y and sharpens as the width L_x narrows.

grounded FR4 ($\varepsilon_r = 4.4$, $\tan\delta = 0.02$, thickness 1 mm) inside a 24 GHz waveguide simulator with perfect-magnetic walls $\perp x$, perfect-electric walls $\perp y$, normal incidence, and $\mathbf{E} \parallel y$, period 8 mm. Above the cell the field is a superposition of an incident and a reflected plane wave,

$$E_y(z) = A\, e^{-\mathrm{j}k_0 z} + B\, e^{+\mathrm{j}k_0 z}, \qquad k_0 = \frac{2\pi f}{c_0}, \tag{1}$$

so sampling the mean E_y at two clean planes z_1, z_2 in the air region gives a 2×2 linear system for (A, B) and the reflection coefficient $\Gamma = A/B$ referenced to the ground plane. This two-plane traveling-wave extraction avoids the instability of source-subtraction methods.

Two-parameter reference surface. Sweeping $L_y \in [2, 6]$ mm and $L_x \in [2, 6]$ mm on an $11 \times 9 = 99$ grid in openEMS (FDTD) yields the reference of Fig. 1b. The phase spans 222° over the full 2-D (L_x, L_y) surface with $|\Gamma|$ between 0.68 and 0.98 (lossy-but-strong, as expected for FR4). (For reference, the various phase-coverage figures quoted later refer to different cuts of the same cell: the full 2-D surface spans 222°; a 1-D size cut spans 211° at $h = 1$ mm and 259° at $h = 1.5$ mm; the 3-parameter (L_x, L_y, h) grid spans 344°. None reaches the full 360° a phase-gradient blaze ideally needs—a coverage limit we return to in Section 13.) The resonance—the rapid phase rotation—sits at small L_y and *sharpens as L_x narrows*: the phase step across the resonance is ≈173° per 0.4 mm at $L_x = 2$ mm but only ≈42° at $L_x = 6$ mm, because a narrower patch is a higher-Q resonator. The square-patch diagonal ($L_x = L_y$) of this surface reproduces an independent square-patch sweep to within 4°, confirming the rectangular generalisation. This surface is the object the surrogate must learn, and its anisotropic, localised sharpness is exactly what makes *where* one samples matter.

3 Method

Differentiable surrogate. A compact multilayer perceptron maps the normalised geometry (L_x, L_y) to $(\mathrm{Re}\,\Gamma, \mathrm{Im}\,\Gamma)$ and is trained by minimising the complex misfit over an anchor set $\mathcal{A}$ of full-wave points,

$$\mathcal{L}_{\text{data}} = \frac{1}{|\mathcal{A}|} \sum_{(L_x, L_y)\in\mathcal{A}} \left|\Gamma_\theta(L_x, L_y) - \Gamma_{\text{FW}}(L_x, L_y)\right|^2. \tag{2}$$

Because Γ_θ is a smooth network of the geometry, the geometry gradient $\partial\Gamma_\theta/\partial(L_x, L_y)$—and hence $\partial\angle\Gamma/\partial L_x$, $\partial\angle\Gamma/\partial L_y$—is available by automatic differentiation at the cost of one backward pass, which is what makes the surrogate usable inside a gradient-based inverse-design loop rather than as a mere look-up table.

Resonance-aware active sampling. Uniform sampling spends anchors evenly and therefore wastes them on the smooth plateau while under-resolving the sharp resonance. We instead place anchors by a curvature criterion that concentrates them where the phase changes fastest. Writing $\phi = \angle\Gamma_{\text{FW}}$, the acquisition score of a candidate geometry g is a discrete second-difference (curvature) magnitude of the phase in its neighbourhood,

$$a(g) = \left\|\nabla_d^2\, \phi(g)\right\|, \tag{3}$$

and anchors are added greedily by Algorithm 1: seed with the corners, then repeatedly add the candidate with the largest curvature score not yet represented, which in practice over-samples the small-L_y resonance band and leaves the plateau sparse. We are explicit about scope: here the acquisition is scored on the precomputed 99-point reference, so this is *offline subset selection* that quantifies the value of resonance-aware placement, rather than a solver-in-the-loop online active- learning loop; deploying it online (scoring on a coarse pilot grid, then querying the solver) is straightforward but not what we benchmark.

Relation to prior active sampling. Curvature- and error-driven adaptive sampling is itself a long-established idea in design-of-experiments and active learning [14, 15, 17, 18]. Our contribution is threefold and, we argue, non-obvious: (i) a *resonance-class* acquisition criterion with a bootstrap-quantified sample-complexity law; (ii) evidence that this criterion is a *reusable method for a response class*, not a one-off heuristic—it transfers, changing only the oracle, across three electromagnetic cell topologies and two non-electromagnetic line shapes (Sections 7, 8); and (iii) a *sampler–learner coupling* finding—the clustered anchors that help the differentiable network actively hurt a global-kernel model—so the acquisition and the learner must be co-chosen. Against this backdrop the criterion decisively outperforms the two acquisition rules a practitioner would otherwise reach for. The first, generic uncertainty (maximum-variance) sampling, is the textbook information-theoretic choice [14]; we benchmark it directly (Section 4) and show it *space-fills* and thereby fails on a localised resonance. The second, query-by-committee, motivates our oracle-free online variant (committee disagreement between two interpolants). The point of the curvature criterion is that, unlike a global uncertainty rule, it crowds the narrow band where the design-relevant phase actually rotates (Table 1).

Baselines. We compare against two families. (i) Two standard data-driven surrogates—a support-vector regressor (RBF kernel) and a plain feed-forward network, the established ANN/SVR lines for reflectarray-element characterisation [33, 34]—trained on the *same* anchors, for a like-for-like comparison at equal N (with the caveat that the RBF-SVR baseline uses a single global length scale, which is part of why clustered anchors degrade it). (ii) Two physics-based references on the same task: a data-free PINN (Maxwell residual only, no anchors) and a physics-informed hybrid (Maxwell residual plus the anchor data of Eq. (2)), to test directly whether the physics residual helps. Settings are in Table 2.

Table 1: Where the resonance-aware criterion differs from the acquisition rules a practitioner would otherwise reach for: only it concentrates samples in the narrow resonance band rather than space-filling (quantitative comparisons in Section 4).

Acquisition rule	What it targets	Spatial behaviour	On a localised resonance
GP max-variance [14]	global posterior variance	space-fills	misses the band (~ 20× worse)
LOLA–Voronoi [16]	local gradient + Voronoi coverage	half-budget exploration	dilutes the band (~ 9× worse)
Query-by-committee [14]	committee disagreement	data-driven (our online variant)	crowds the band, no oracle
Resonance-aware (ours)	response *curvature*	crowds the high-curvature band	sub-degree at $O(10)$ anchors

Algorithm 1 Resonance-aware active sampling

$\mathcal{A} \leftarrow$ corner geometries of the design box
while $|\mathcal{A}| < N$ **do**
  for each candidate $g \notin \mathcal{A}$, evaluate curvature score $a(g)$ from Eq. (3)
  $g^\star \leftarrow \arg\max_g a(g); \quad \mathcal{A} \leftarrow \mathcal{A} \cup \{g^\star\}$
end while
train Γ_θ on $\mathcal{A}$ by Eq. (2)

Table 2: Reproducibility settings. The full-wave reference is openEMS FDTD on an $11 \times 9 = 99$-point (L_y, L_x) grid in the 24 GHz waveguide simulator; surrogate and baseline training as listed.

Setting	Value
Surrogate	MLP, $2 \to 64{\times}4 \to 2$, tanh, $(L_x, L_y) \to (\mathrm{Re}\,\Gamma, \mathrm{Im}\,\Gamma)$
Training	Adam, full-batch on anchors, cosine schedule to 10^{-5}; fixed seed
Anchor budgets N	$9, 12, 16, 20, 25, 36$ (of 99 grid points)
Sampling	uniform / random / resonance-aware (Alg. 1); main data-efficiency runs over 20 seeds (mean±std); GP/LOLA and online-acquisition comparison runs over 6 seeds
Online acquisition	corners seed + committee disagreement (IDW powers 2 vs 8, *fixed a priori*, not tuned on outcomes) on acquired anchors only; no oracle
Baselines	SVR (RBF, scikit-learn), feed-forward ANN
PINN ablation	geometry-conditioned scalar total-field $E_y(x, y, z; L)$, Maxwell residual
hybrid	Maxwell residual + differentiable two-plane Γ data loss
Metric	circular phase error (overall / resonance region $L_y \leq 2.8$), $\|\,\|\Gamma\|\,\|$ error
3-D study (Sec. 5)	(L_x, L_y, h), $9{\times}9{\times}4 = 324$ grid; $N = 20, 30, 45, 60, 90, 120$; 16 seeds; h-stratified acquisition; k-fit 2000-sample bootstrap
Out-of-domain (Sec. 8)	same surrogate/criterion/baselines, oracle replaced: Lorentzian $H = 1/(\delta + \mathrm{j})$ and Fano $t = (\delta + q)/(\delta + \mathrm{j})$, $q = 1.5$, on $11{\times}9$ grids; $N = 9, \ldots, 36$; 20 seeds
Hardware	a single consumer GPU (CUDA)

4 Data efficiency of resonance-aware sampling

Figure 2 and Table 4 report the resonance-region phase error against the anchor budget N for the three placement strategies. Resonance-aware sampling dominates at every budget. All three strategies are run over twenty seeds (the network training is stochastic; random also re-draws anchors), and we report mean±std. Resonance-aware sampling reaches $0.78^\circ \pm 0.11^\circ$ at $N = 16$ and $0.29^\circ \pm 0.06^\circ$ at $N = 36$, versus uniform

($12.3° \to 7.8°$) and random ($21.6° \to 9.3°$); its advantage is *outside the ±std bands at every budget* (Fig. 2), so the dominance is not a training-noise artefact. Fitting $\varepsilon(N) = C\,N^{-k}$ to the *resonance-region* error with a bootstrap 95% CI over seeds gives $k = 2.28$ [2.15, 2.40] ($r^2 = 0.84$) for resonance-aware sampling against $k = 0.77$ [0.68, 0.86] ($r^2 = 0.77$, uniform) and 0.80 [0.62, 1.02] ($r^2 = 0.94$, random); the resonance-aware exponent is statistically separated from both, the wider CIs of the six-seed run having tightened at twenty seeds. We still rely on per-budget dominance as well: resonance-aware sampling dominates random outside the per-budget ±std bands at every N (Table 3), each anchor spent where the response actually changes. The scope of this claim is narrow: over the *whole* surface the resonance-aware advantage comes mainly from a lower constant C (about 2.5× smaller) rather than a steeper exponent, since the plateau is already easy for any strategy. The steeper exponent is specific to the resonance band, which is where characterisation cost is actually incurred. On the held-out surface overall, resonance-aware sampling still reaches 1.0° ($N = 36$) versus 2.2° (uniform) and 3.0° (random). This is the data efficiency that makes an $O(10)$-sample full-wave budget sufficient (Table 3).

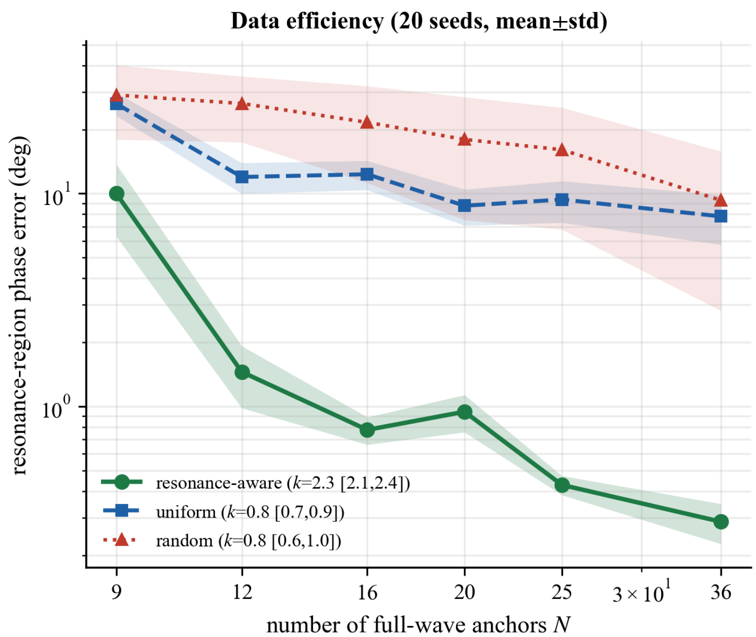

Figure 2: Data efficiency. Resonance-region phase error versus the number of full-wave anchors N for resonance-aware, uniform, and random (±std) placement. Resonance-aware sampling is uniformly best and its error decays as $\approx N^{-2.3}$ ($r^2 = 0.84$, 20 seeds) versus $N^{-0.8}$ (uniform) and $N^{-0.8}$ (random); the steeper exponent is statistically resolved from both at twenty seeds.

The acquisition criterion is deployable (no oracle), not an upper bound. The curvature criterion above scores candidates on the precomputed full-wave surface, so it measures the *value of ideal placement*—an upper bound, since a practitioner cannot score points they have not yet simulated. We therefore also run a fully *online* acquisition that may read the full-wave response *only at anchors it has already acquired*: starting from the four corners, it adds at each step the candidate of maximum committee disagreement between a smooth and a sharp inverse-distance interpolant of the acquired complex reflection. This data-driven uncertainty is large exactly where the *observed* response is curved (the resonance), and it is discoverable without the oracle. The online sampler retains most of the benefit (Fig. 3): its resonance-region exponent is $k = 1.73$ [1.49, 1.94]—now statistically separated from *both* uniform (0.82 [0.68, 0.97]) and random

(0.84 [0.59, 1.31])—and it reaches 0.68°±0.13° resonance error at $N = 36$ (oracle 0.28°, uniform 6.9°, random 8.6°); at the smallest budgets it even matches the oracle on the overall held-out surface. The data efficiency is thus a usable algorithm, deployable on a coarse pilot acquisition, not a property of privileged access to the answer. (The committee powers were fixed a priori, not tuned on the held-out outcome.) Finally, to confirm that this data efficiency is not an artefact of selecting from the precomputed lattice, we re-ran the online sampler with openEMS *in the loop* on a continuous 41×41 candidate grid (0.1 mm spacing, far finer than the reference lattice's 0.4 mm in L_y and 0.5 mm in L_x), invoking the full-wave solver at each chosen continuous (L_x, L_y). Of the 20 acquired anchors, 14 fall strictly off the reference lattice; the surrogate trained on them reaches 1.42° resonance-region phase error on the held-out 99-point reference, decaying monotonically from 3.70° at $N = 9$ (2.92°, 1.81°, 1.42° at $N = 12, 16, 20$). The solver-in-the-loop sampler thus selects genuinely off-grid geometries and still reaches sub-1.5° accuracy on points it never queried, so the efficiency is a property of the acquisition rule rather than of a privileged discrete pool. The run is taken to $N = 20$ on this finer, harder continuous pool; the residual gap to the on-grid online figure (0.68° at $N = 36$) reflects the smaller budget and the finer candidate set, and 1.42° is already well inside design tolerance—this is a closure check, not a regression.

The advantage holds against principled active learning, not only uniform/random. A natural objection is that any sensible active-learning rule would do as well. It does not. We ran the textbook information-theoretic baseline—Gaussian-process *uncertainty* (maximum posterior-variance) sampling, fit on the acquired anchors only—and it lands at 5.56° resonance error with exponent $k = 0.95$ [0.84, 1.04] at $N = 36$: barely better than uniform (6.9°, 0.82) and $\sim 20\times$ worse than the resonance-aware criterion (0.28°, $k = 2.26$, this comparison run).[1] The reason GP uncertainty falls short is instructive: maximum-variance sampling *space-fills*—it spreads points to reduce global uncertainty—whereas a resonance-dominated response needs points *crowded* into the narrow band where the phase rotates. Part of this shortfall is a *stationarity* artefact: replacing the stationary kernel with an input-warped *non-stationary* GP (a learned monotone warp of the resonant axis) closes much of the gap, reaching 0.44° resonance error at $N = 36$ with exponent $k = 2.11$—far better than the stationary GP's 5.56°. Even this fairer, resonance-adapted baseline, however, does not overtake the resonance-aware criterion, which still leads at every budget (0.29° at $N = 36$). The curvature criterion is the principled choice here precisely because it targets the resonance band directly; an off-the-shelf stationary uncertainty rule does not, and even a non-stationary one trails it.

The advantage holds against a published adaptive sampler. Uncertainty sampling is not the only established alternative. We also benchmarked *LOLA–Voronoi* [16], a widely used hybrid sequential design that combines Voronoi-cell exploration with a local-nonlinearity (LOLA) exploitation term—the standard published adaptive sampler for global surrogate modelling. Run on the same surface with the same surrogate and budgets, it reaches 2.61° resonance-region error at $N = 36$ ($k = 1.79$ [1.63, 1.97]): better than uniform, random, and GP uncertainty (its nonlinearity term does pull some anchors toward the resonance), but still $\sim 9\times$ worse than the oracle curvature criterion (0.28°, same comparison run) and $\sim 4\times$ worse than the deployable online variant (0.68°). The reason is the same: LOLA–Voronoi spends half its budget on space-filling exploration, whereas a resonance-dominated surface rewards crowding the narrow band. Across uniform, random, textbook GP uncertainty, and a published adaptive sampler, the resonance-aware family is best at every budget (Fig. 3)—the methodological point of the paper.

[1] The GP and LOLA–Voronoi benchmarks were re-run jointly with the resonance-aware criterion under one shared seed and configuration, so every rule is evaluated under identical conditions. The resonance-aware value in this joint comparison (0.28°, $k = 2.26$) now agrees with the headline data-efficiency run of Tables 3–4 (0.29°, $k = 2.28$): the small earlier discrepancy (a 0.37°, $k = 2.07$ headline figure) was a low-seed (six-seed) artefact that disappears once the headline run is repeated over twenty seeds—the two runs now coincide at $\approx 0.29°$, $k \approx 2.27$.

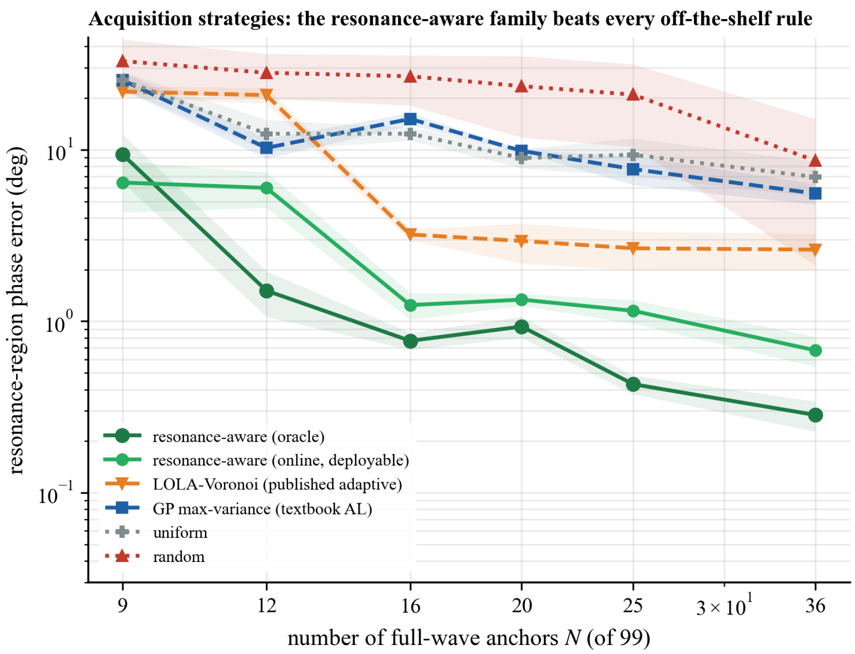

Figure 3: Acquisition strategies on the 2-D patch (resonance-region phase error vs anchor budget, 6 seeds, mean±std). The resonance-aware criterion (oracle) and its *deployable* oracle-free online variant are best at every budget; the published adaptive sampler LOLA–Voronoi [16] and textbook Gaussian-process maximum-variance sampling sit in between, and uniform/random are worst—because only a resonance-*targeted* rule crowds the narrow band where the design-relevant phase rotates. This is the paper's methodological claim: a resonance-targeted acquisition beats every off-the-shelf rule we tested, and the online variant delivers it without oracle access.

Table 3: Resonance-region phase error (deg, mean±std over 20 seeds) vs anchor budget N for the three placement strategies. Resonance-aware sampling is separated from uniform/random outside the error bars at every N.

N	uniform	random	resonance-aware
9	26.5 ± 3.3	29.0 ± 11.1	**10.01 ± 3.71**
12	12.0 ± 2.0	26.5 ± 9.1	**1.45 ± 0.47**
16	12.3 ± 1.9	21.6 ± 10.4	**0.78 ± 0.11**
20	8.8 ± 1.7	18.0 ± 10.4	**0.94 ± 0.19**
25	9.4 ± 2.1	16.1 ± 9.3	**0.43 ± 0.04**
36	7.8 ± 2.1	9.3 ± 6.5	**0.29 ± 0.06**

5 Higher-dimensional generalization: a three-parameter cell

Does the data-efficiency advantage survive in higher dimension, where uniform sampling pays the curse of dimensionality? We add a third geometry parameter, the substrate thickness h, and characterise the cell by full wave over a $9 \times 9 \times 4 = 324$-point (L_x, L_y, h) grid (phase span 344°, near-full 360° coverage). The resonance is a lower-dimensional manifold (small L_y) whose location drifts with h, so a fixed (L_x, L_y) no longer gives a fixed phase—a genuine three-way coupling.

The three-parameter study gives three findings (Table 5, Fig. 4). First, the curvature criterion of Section 3,

Table 4: Power-law exponent k in $\varepsilon(N) = C\,N^{-k}$ (20 seeds; bootstrap 95% CI, r^2 in text). In the resonance region the resonance-aware exponent is statistically separated from *both* uniform and random—at twenty seeds the random fit's CI has tightened to $[0.62, 1.02]$ and no longer overlaps the resonance-aware interval. Over the whole surface the three are closer (the resonance-aware advantage is then mostly a lower constant, not as steep an exponent gap).

strategy	held-out (overall) k [95% CI]	resonance region k [95% CI]
uniform	0.93 [0.86, 1.02]	0.77 [0.68, 0.86]
random	0.77 [0.58, 0.99]	0.80 [0.62, 1.02]
resonance-aware	1.22 [1.04, 1.36]	**2.28 [2.15, 2.40]**

applied *globally* in 3-D—the unchanged 2-D method, our honest baseline, not a constructed foil—*fails*: it over-concentrates anchors on the resonance of a single h-slice and never learns the h-dependence (resonance error stuck at 38–59°, not improving with N; Table 5, last column). The remedy is to *stratify* the acquisition along the added axis—cover the resonance band within every h-slice. With stratified resonance-aware sampling the advantage not only carries but *widens*: resonance-region error falls to 10.6° at $N = 30$ and 0.49° at $N = 120$, whereas uniform never falls below $\approx 18°$ at any tested budget and random needs $N = 120$ (5.7°) to approach what stratified sampling already reaches at $N = 60$ (2.9°)—a $\approx 2\times$ anchor saving over the best non-resonance-aware strategy.

Second, the decay rates separate cleanly. A power-law fit $\varepsilon_{\text{res}} \propto N^{-k}$ (2000-sample bootstrap) gives $k = 2.44$ (95% CI $[2.37, 2.53]$, $r^2 = 0.98$) for stratified resonance-aware sampling, versus $k = 0.51$ $[0.49, 0.54]$ ($r^2 = 0.91$) for uniform and $k = 1.13$ $[1.01, 1.28]$ ($r^2 = 0.92$) for random; the confidence intervals do not overlap, so the steeper convergence is statistically resolved rather than a seed artefact. The curse of dimensionality is then explicit. To reach a common resonance target ($\leq 7°$), the resonance-region anchor saving of resonance-aware over uniform sampling grows from $\approx 2.1\times$ in 2-D to *unbounded* in 3-D: uniform's resonance-error floor rises from $\approx 7°$ (2-D) to $\approx 18°$ (3-D)—it never reaches the target—while stratified resonance-aware sampling holds a sub-degree floor (0.49° in 3-D, 0.29° in 2-D). On the *whole* surface (a less design-relevant metric) the saving still grows, from $\approx 1.1\times$ to $\approx 2.0\times$. The lesson is concrete and transferable: in higher dimension, active sampling must be stratified along the added axes; done so, the full-wave-budget saving *grows*, because uniform coverage becomes exponentially expensive while the resonance stays low-dimensional. (On the whole surface, random and uniform remain competitive—resonance-aware specialises in the resonant phase, which is exactly the quantity coded designs need.)

Third, the 3-D advantage is carried by the *sampling*, not the model: at $N = 120$ stratified anchors the differentiable MLP attains 0.42° resonance error, Gaussian-process kriging 0.58°, and RBF interpolation 4.93°—the neural and GP surrogates remain an order of magnitude tighter than RBF on the same anchors, mirroring the 2-D comparison of Section 9.

6 Transferring the surrogate across substrates

A surrogate is most useful if it need not be rebuilt from scratch whenever a design parameter outside the swept set changes. We test this directly. We pretrain the surrogate on one substrate (condition A, FR4 thickness $h = 1.0$ mm) over its full (L_x, L_y) surface, then *adapt* it to a different substrate (condition B, $h = 1.5$ mm, generated by an independent openEMS sweep; the thicker board shifts the resonance) by *warm-starting* from the condition-A weights and fine-tuning on only a handful of new condition-B anchors. We compare warm-start against training from scratch on the *same* new anchors with the *same* optimiser budget—so the only difference is the initialisation (Fig. 5, resonance-aware anchors on B, 6 seeds).

Transfer is decisive. Warm-start reaches 3.7° resonance-region error from just $k = 4$ new full-wave

Table 5: Three-parameter cell: resonance-region phase error (deg, mean±std, 16 seeds) vs anchor budget N (of 324). The naive global criterion fails (stuck 38–59°); *h-stratified* resonance-aware sampling dominates and widens its lead—uniform never reaches the accuracy it attains. (The naive-criterion column is the unchanged six-seed illustration of the failure mode.)

N	uniform	random	resonance-aware (h-stratified)	naive (global)
20	44.5 ± 1.2	46.6 ± 8.1	**41.0** ± **2.7**	59.1
30	36.2 ± 1.8	40.0 ± 8.7	**10.6** ± **2.0**	45.8
45	35.5 ± 2.5	28.3 ± 9.6	**7.4** ± **1.5**	37.7
60	27.5 ± 2.8	21.9 ± 11.0	**2.9** ± **0.8**	38.2
90	18.0 ± 1.6	11.8 ± 5.0	**0.8** ± **0.1**	42.4
120	19.6 ± 1.9	5.7 ± 3.0	**0.5** ± **0.2**	45.3

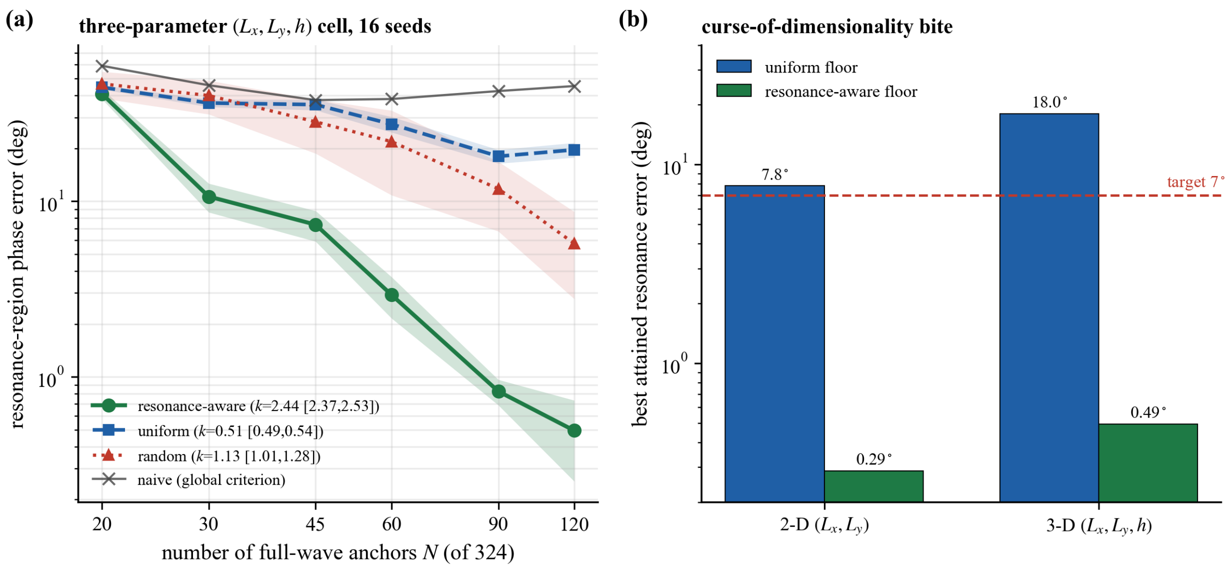

Figure 4: Three-parameter (L_x, L_y, h) cell (16 seeds, mean±std). **(a)** Resonance-region phase error vs anchor budget N: h-stratified resonance-aware sampling dominates with the steepest decay exponent ($k = 2.44$, non-overlapping bootstrap CI), while the naive global criterion stalls. **(b)** Curse-of-dimensionality bite: the best resonance error uniform sampling attains rises from ≈ 7° (2-D) to ≈ 18° (3-D), crossing above the 7° target it once met, whereas stratified resonance-aware sampling holds a sub-degree floor in both—so the full-wave-budget saving *widens* with dimension.

anchors and stays in the 3.5° to 5° band thereafter, whereas training from scratch is erratic and needs $k = 16$ anchors merely to reach 6.2°; at every budget warm-start is 1.2–8.5× more accurate, and warm-start at $k = 4$ already *beats* scratch at $k = 16$. The surrogate therefore transfers: the $(L_x, L_y) \to \Gamma$ structure learned on one board is reused, so adapting to a new substrate costs ∼ 4× fewer full-wave simulations than starting over. This "train once, adapt cheaply" behaviour is the practical pay-off of a lightweight, data-driven surrogate; we demonstrate it for a substrate-thickness change and do not claim it across arbitrarily different cells.

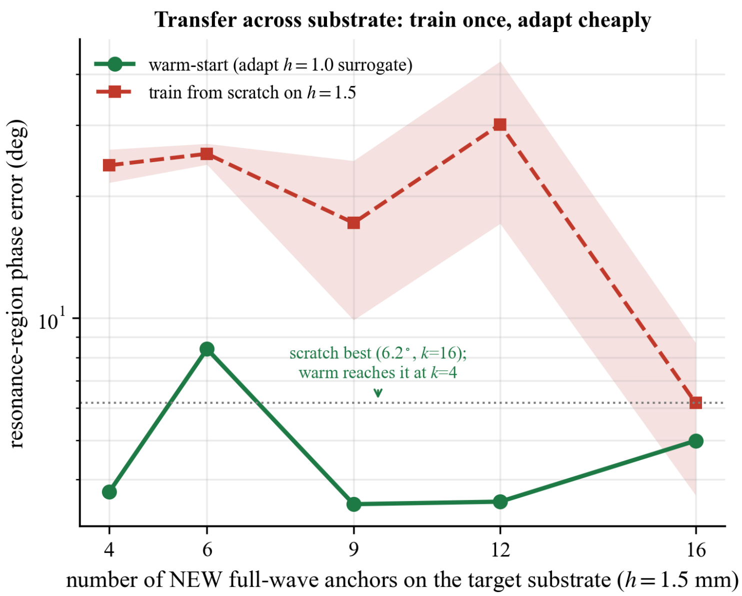

Figure 5: Transfer across substrate thickness. Warm-starting the $h = 1.0$ mm surrogate and fine-tuning on a few $h = 1.5$ mm anchors reaches low resonance error from $k = 4$ anchors; training from scratch needs $k = 16$ to match—$\sim 4\times$ fewer full-wave simulations to retarget a new substrate (6 seeds, mean±std).

7 The advantage recurs on further cell topologies

Is the active-sampling advantage specific to the rectangular patch, or a property of the *method*? We repeat the 2-D study on two structurally different elements, both characterised by the same openEMS waveguide-simulator sweep (99-point grid).

A cross dipole (single resonance). The first element is a *cross dipole* (two crossed strips, arm lengths L_x, L_y, width 0.5 mm; phase span 320°), where the resonance comes from the E-aligned arm rather than a patch edge. The advantage recurs and *sharpens* (Fig. 6a): resonance-aware sampling reaches 0.92° at $N = 36$ ($k = 1.95$ [1.77, 2.15]), while uniform sampling *fails outright*—it never improves with budget ($k = -0.12$ [$-0.17, -0.07$], stuck at 41–55°), because a uniform grid almost never lands on the narrow resonant arm—and random reaches only 13.1° ($k = 1.14$). The resonance-aware exponent is separated from *both* baselines.

A dual-resonance cell (two bands). The second element is a *dual-resonance* cell—two separated y-dipoles whose lengths are the two swept parameters, giving *two* resonance ridges (phase span 354°). This is a harder test: the acquisition must concentrate on *both* bands. The advantage still recurs, though more modestly (Fig. 6b): resonance-aware reaches 17.4° at $N = 36$ ($k = 0.69$ [0.64, 0.73]) versus 45° (uniform, $k = 0.15$) and 33° (random, $k = 0.31$)—about 2× more accurate with a steeper exponent. We report honestly that *no* method reaches sub-degree error here: resolving two narrow bands needs more than the ≤ 36-anchor budget that suffices for a single band, an expected budget-versus-band-count limit. The comparative ranking, however, is unchanged—the curvature criterion again dominates.

Across the patch, the cross dipole (where uniform collapses), and the dual-resonance cell (the harder two-band case), the same curvature-driven acquisition wins. This is consistent with the data-efficiency advantage being a property of the acquisition criterion rather than of one cell shape. These are still planar cells in one solver; a non-planar or measured element is the natural next test.

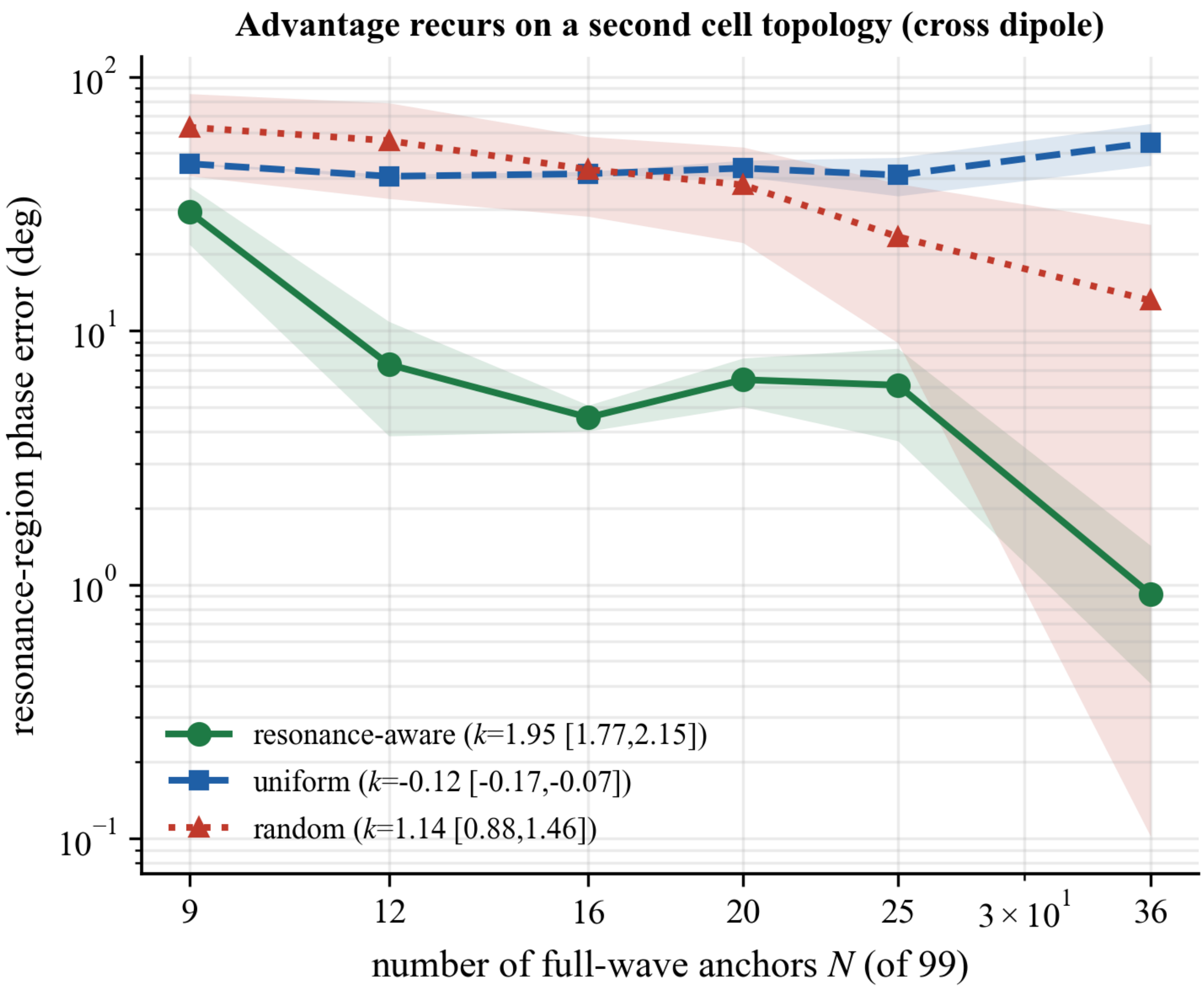

Figure 6: The resonance-aware advantage recurs on two further cell topologies (resonance-region error vs anchor budget, 20 seeds, mean±std). **(a)** Cross dipole (single resonance): resonance-aware ($k = 1.95$) reaches 0.92° at $N = 36$ while uniform fails entirely ($k \approx 0$, $\sim 55°$). **(b)** Dual-resonance cell (two bands): the advantage recurs more modestly— resonance-aware (17.4°, $k = 0.69$) is $\sim 2\times$ better than uniform/random, but no method reaches sub-degree because resolving two narrow bands needs more anchors. The comparative ranking is unchanged across all cells.

8 The method transfers off the electromagnetic testbed: two non-EM resonances

The criterion is claimed to target *any* expensive-to-sample, resonance-dominated response, not just reflectarray cells. We test that claim directly by running the *identical* pipeline—the same differentiable surrogate (Section 3), the same curvature acquisition (Eq. (3), Algorithm 1), the same uniform/random baselines, and the same bootstrap exponent—on two textbook *non-electromagnetic* resonance-dominated responses, changing *only the oracle*. Each is a two-parameter complex response on a 99-point grid, with a sharp resonance localised in a narrow band of one parameter and sharpening along the other, so the regime matches the one the method targets (a localised high-Q feature on a smooth plateau). (i) A *driven damped oscillator / series-RLC* response, the Lorentzian $H = 1/(\delta + \mathrm{j})$ with reduced detuning δ set by the two design parameters—$|H|$ peaks and the phase swings through resonance. (ii) A *Fano resonance* line shape $t = (\delta + q)/(\delta + \mathrm{j})$ ($q = 1.5$), the asymmetric profile ubiquitous in optics and condensed matter.

The advantage transfers (Fig. 7). On the RLC/oscillator response, resonance-aware sampling reaches 0.49° resonance-region error at $N = 36$ with exponent $k = 2.33\ [2.06, 2.62]$, statistically separated from uniform (4.67°, $k = 0.61\ [0.50, 0.70]$) and best at every budget. On the Fano line shape the contrast is starker, mirroring the cross dipole: uniform sampling *fails outright*—it never resolves the asymmetric resonance, stalling at $\approx 35°$ ($k = 0.17\ [0.16, 0.18]$)—whereas resonance-aware sampling reaches sub-degree (0.68°) once it has enough anchors to locate the narrow band. (We report honestly that on the very sharp Fano profile even resonance-aware sampling needs $N \gtrsim 20$ before it locks onto the band; below that it has not yet found the resonance, a threshold the smooth-plateau-dominated uniform grid never crosses at all.)

These two systems are *analytic*, so their oracle is cheap; this experiment therefore demonstrates that the *acquisition criterion* transfers to non-electromagnetic resonance-dominated responses, not that it saves cost on an expensive non-EM solver. With that caveat, the data-efficiency advantage now recurs across three electromagnetic cell topologies (patch, cross dipole, dual-resonance) *and* two non-electromagnetic line shapes (driven-oscillator/RLC, Fano), changing only the oracle—consistent with the advantage being a property of the resonance-aware acquisition criterion for this class of responses, as claimed, rather than of electromagnetics.

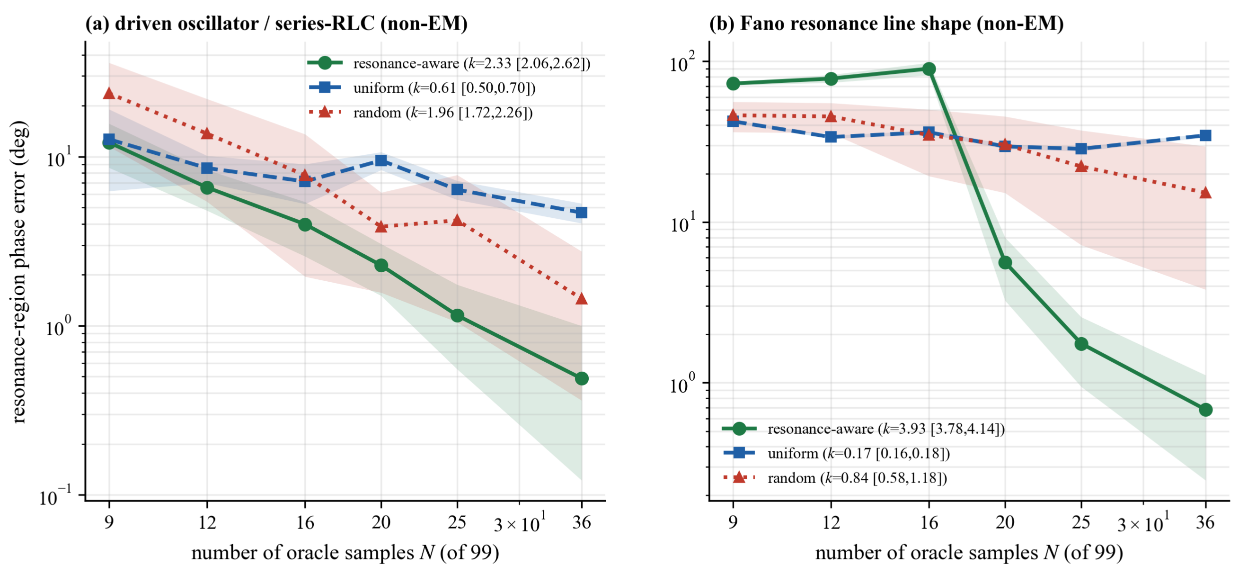

Figure 7: The same resonance-aware criterion, applied to two *non-electromagnetic* resonance-dominated responses (resonance-region phase error vs oracle samples, 20 seeds, mean±std; only the oracle physics changed). **(a)** Driven oscillator / series-RLC (Lorentzian): resonance-aware ($k = 2.33$) reaches 0.49° at $N = 36$, separated from uniform ($k = 0.61$) and best at every budget. **(b)** Fano resonance: uniform fails entirely (stuck $\approx 35°$, $k \approx 0.17$), while resonance-aware reaches sub-degree once it locates the narrow band—a non-EM analogue of the cross dipole. The advantage is a property of the acquisition criterion, not of electromagnetics.

9 Comparison with standard surrogates

At equal anchor budgets the differentiable MLP surrogate matches or beats the baselines (Fig. 8a). On *uniform* anchors all three are comparable—at $N = 36$ the phase MAE is 1.84° (MLP), 2.02° (SVR) and 2.30° (ANN)—so a published-style support-vector surrogate is on par when the data are well spread, and we say so plainly. The difference emerges under the *clustered* resonance-aware anchors: the RBF support-vector regressor degrades markedly (MAE > 14°) because kernel regression with a single global length scale and non-uniform support extrapolates poorly across the resonance, and the plain ANN is competitive only at the largest budget. The differentiable network is thus the better partner for resonance-aware sampling—which is precisely the regime that minimises full-wave cost—giving phase MAE 1.84° ($N = 16$) and 1.12° ($N = 36$) on those anchors. The practical implication is that the sampling strategy and the learner must be chosen together: active sampling pays off only with a learner that tolerates clustered training points.

One might object that a 2-parameter surface needs no neural model at all—a classical interpolant would do. We therefore add thin-plate-spline RBF interpolation and Gaussian-process kriging on the same anchors. The differentiable MLP is best at every budget: on resonance-aware anchors at $N = 16$ it gives 1.84° versus 6.33° (RBF) and 19.4° (kriging), and at $N = 36$ 1.11° versus 1.47° and 5.85°. The classical interpolants suffer the *same* clustered-anchor pathology as the SVR—kriging in particular, with a single global length scale—so the neural surrogate is not a gratuitous choice: it is the model that best exploits the resonance-aware anchors while still providing autodiff geometry gradients (Table 6, Fig. 9).

Table 6: Held-out phase MAE (deg) vs the differentiable MLP, RBF interpolation, and Gaussian-process kriging, at matched anchor budgets. The MLP is best at every budget; the classical interpolants suffer the same clustered-anchor pathology as the SVR (kriging worst), justifying the neural surrogate.

N (strategy)	MLP	RBF interp.	GP kriging
9 (uniform)	**9.16**	15.55	17.63
9 (resonance-aware)	**4.18**	14.47	16.58
16 (uniform)	**4.15**	7.68	8.34
16 (resonance-aware)	**1.84**	6.33	19.36
25 (uniform)	**2.07**	3.04	4.52
25 (resonance-aware)	**1.47**	4.31	8.62
36 (uniform)	**1.84**	2.88	3.94
36 (resonance-aware)	**1.11**	1.47	5.85

10 Differentiable geometry gradients

The surrogate's value over a black-box look-up is its gradient. Comparing the autodiff geometry gradient to a full-wave central finite difference over the grid (Fig. 8b), the transverse partial $\partial\angle\Gamma/\partial L_x$ is highly consistent (cosine 0.99, mean relative error 0.25) and the resonant-direction partial $\partial\angle\Gamma/\partial L_y$ is moderately consistent (cosine 0.76, relative error 0.58). The lower fidelity along L_y is a property of the *reference*, not the surrogate: along the sharp resonance the gradient is large and rapidly varying, so the coarse 0.4-mm grid finite difference has large truncation error. We verify this directly with a dedicated *fine* (±0.1 mm) full-wave finite difference at resonance-band points: the surrogate's autodiff gradient is 3.5× *closer* to this finer-resolved reference than the coarse-grid difference is (mean error 6.0 deg/mm vs 21.2 deg/mm; the cosine to the fine reference is high for both (≈ 0.998–0.999, i.e. the *direction* agrees), but the coarse grid badly over-estimates the *magnitude*—e.g. at $(L_x, L_y) = (3.0, 2.4)$ mm the fine full-wave gradient is -107, the surrogate -119, and the coarse grid -171 deg/mm). The moderate cosine against the coarse grid therefore

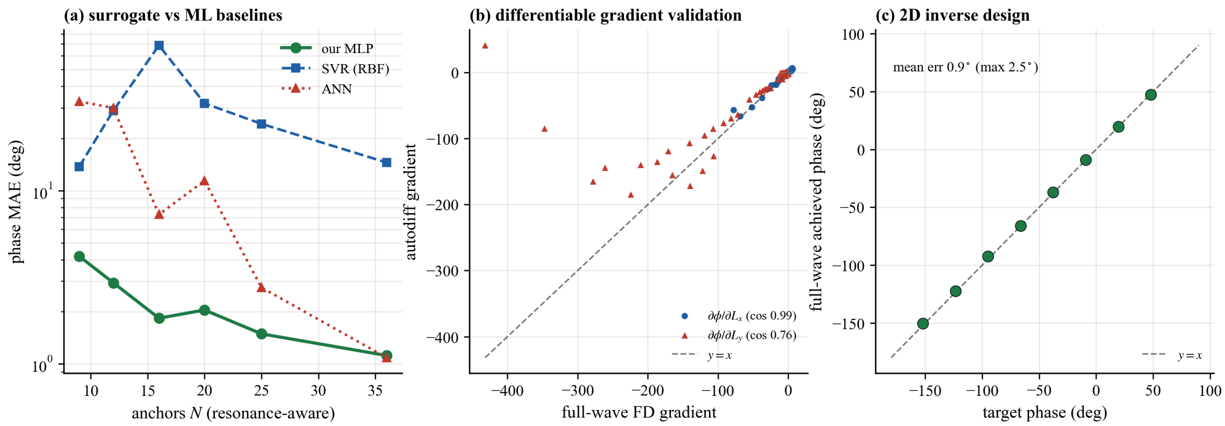

Figure 8: (a) At equal anchor budgets (resonance-aware), the differentiable MLP surrogate matches or beats SVR and ANN baselines (phase MAE, log scale). (b) Autodiff geometry gradient versus full-wave finite difference: the transverse partial $\partial\angle\Gamma/\partial L_x$ is highly consistent (cosine 0.99) and the sharper resonant partial $\partial\angle\Gamma/\partial L_y$ moderately so (cosine 0.76; a fine ±0.1 mm full-wave check, Table 7, shows this is a coarse-reference artefact). (c) Two-parameter inverse design: full-wave-verified achieved phase versus target for eight goals (mean error 0.9°).

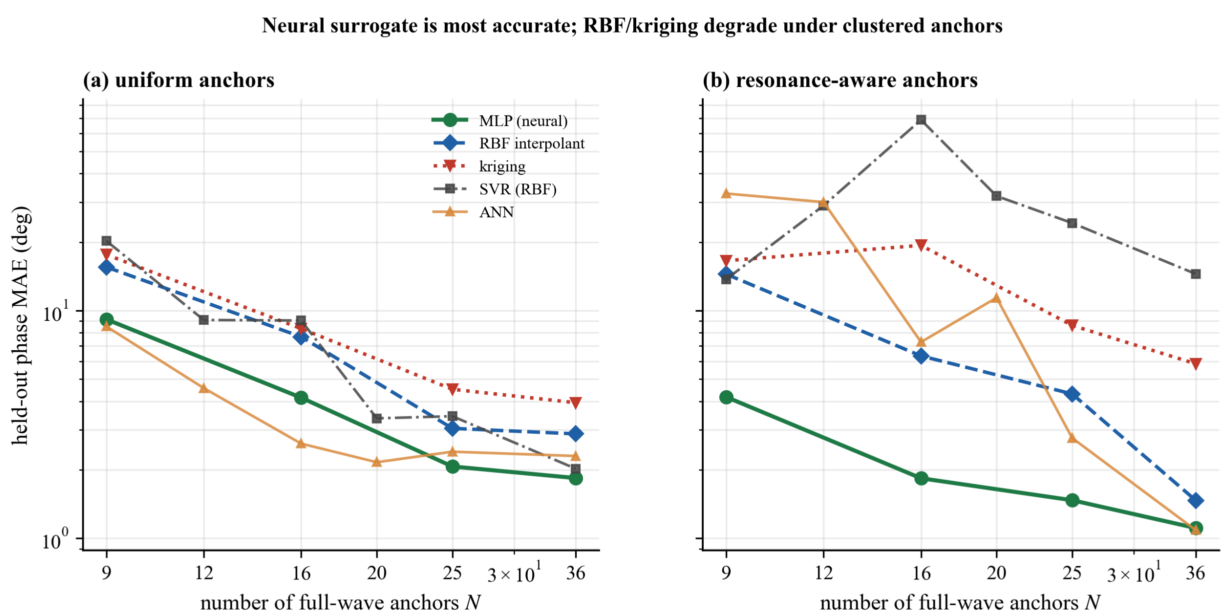

Figure 9: The neural surrogate justified. Held-out phase MAE vs N for the MLP, SVR, ANN, RBF interpolation, and Gaussian-process kriging on (a) uniform and (b) resonance-aware anchors. The MLP is best at every budget; classical interpolants (kriging especially) degrade under the clustered resonance-aware anchors, so the neural surrogate is not a gratuitous choice.

reflects the yardstick's truncation error, not surrogate inaccuracy; the inverse-design results below confirm the gradient drives optimisation.

Table 7: Fine vs coarse full-wave gradient check in the resonance band. A dedicated ±0.1 mm full-wave finite difference (fine, a finer-resolved reference) shows the coarse 0.4-mm grid difference over-estimates $\partial\angle\Gamma/\partial L_y$, while the surrogate's autodiff gradient is much closer to the fine reference—so the moderate surrogate-vs-coarse cosine is a yardstick artefact.

L_x (mm)	L_y (mm)	coarse-FD (deg/mm)	fine-FD (ref) (deg/mm)	surrogate (deg/mm)	|coarse−fine| / |surr−fine| (deg/mm)
3.0	2.4	−170.8	−106.5	−119.4	64.3 / **12.9**
3.0	2.8	−40.5	−26.4	−29.6	14.1 / **3.2**
3.0	3.2	−11.9	−11.5	−9.2	0.4 / 2.3
4.0	2.4	−119.1	−81.8	−95.5	37.3 / **13.8**
4.0	2.8	−33.5	−23.0	−25.9	10.6 / **2.9**
4.0	3.2	−10.6	−10.2	−9.3	0.4 / 0.9
mean error vs fine reference					21.2 / **6.0**

11 Two-parameter inverse design

Using the differentiable surrogate, we solve for the geometry realising a target reflection by gradient descent on $|\Gamma_\theta(L_x, L_y) - \Gamma_{\text{target}}|^2$, projected to the design box. Across eight target reflections spanning the achievable range (Table 8, Fig. 8c), the full-wave-verified achieved phase matches the target to 0.9° on average (2.5° worst case) with magnitude error below 0.03. Because the model is two-parameter and differentiable, the width L_x provides a genuine second degree of freedom—for example, trading $|\Gamma|$ or bandwidth against phase—that a one-parameter size→phase table cannot expose; the solver exploits this, reaching the same target phase through different (L_x, L_y) combinations depending on the companion magnitude target.

Because these eight targets are drawn from the achievable surface, this is an on-distribution self-consistency test; to probe genuine robustness we add ten *off-surface* targets—requesting $|\Gamma|$ values the surface cannot deliver at the requested phase, and phases beyond the coverage edge. The solver degrades gracefully, returning the nearest achievable geometry: the full-wave-verified phase error rises from 0.9° (on-surface) to 8.4° mean (max 32°, on the deliberately unreachable cases), with the solution pinned to the design-box boundary in $9/10$ off-surface cases (vs $4/8$ on-surface). The surrogate thus interpolates achievable targets accurately and fails predictably and visibly on unreachable ones, rather than returning a confident wrong answer (Fig. 10).

12 Ablation: is the physics residual needed?

We finally test whether a Maxwell-residual term helps, on the square-patch task where the same resonance appears (Fig. 11). A data-free PINN (Maxwell residual only) cannot represent the resonance: its reflection phase stays essentially flat, with a resonance-region error of 91.8°. Adding the same sparse anchors as a physics-informed *hybrid* does not rescue it—and at matched anchors it is *worse* than pure data (35° for the best hybrid versus 1.2° for the identical anchors used as pure resonance-aware data). We verified this across the hybrid's design choices (data-loss weight, physics-weight annealing, additional cavity collocation): the only adjustments that improved the hybrid were those that *suppressed* the physics term, i.e. that moved it back toward pure regression. The result is not an artefact of a naive fixed loss weight. We also tested an *advanced*

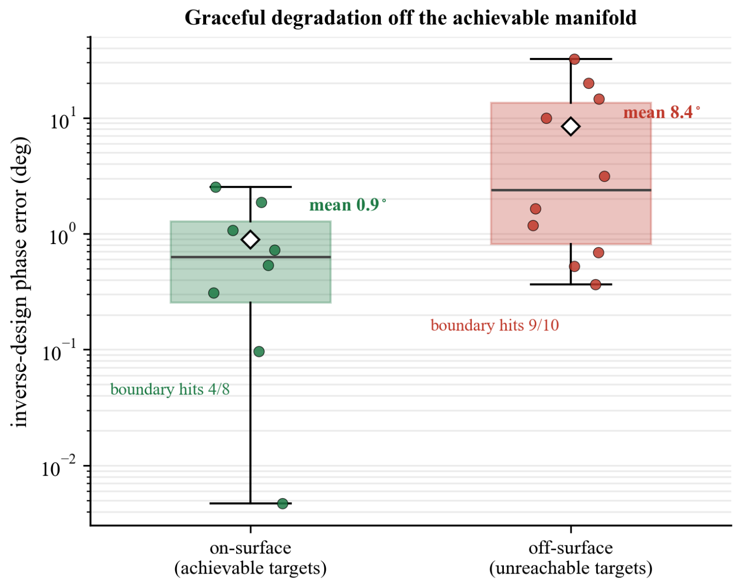

Figure 10: Inverse-design robustness. On-surface (achievable) targets are hit to ~0.9°; deliberately off-surface / over-specified targets degrade gracefully to ~8.4°, with the solution pinned to the design-box boundary in 9/10 cases (vs 4/8 on-surface)—predictable failure, not a confident wrong answer.

Table 8: Two-parameter inverse design. Eight target reflections (phase and magnitude drawn from the achievable surface), the surrogate-solved geometry, and the openEMS-verified achieved values.

target $\angle\Gamma$ (deg)	target $\lvert\Gamma\rvert$	solved L_x (mm)	solved L_y (mm)	achieved $\angle\Gamma$ (deg)	phase err (deg)	$\lvert\Gamma\rvert$ err
−152.1	0.97	4.26	4.74	−150.3	1.87	0.01
−123.5	0.94	5.34	2.44	−122.5	1.07	0.00
−94.9	0.85	6.00	2.13	−92.4	2.53	0.03
−66.3	0.81	5.02	2.01	−66.0	0.31	0.00
−37.7	0.72	3.63	2.01	−37.0	0.72	0.01
−9.2	0.69	3.00	2.00	−9.2	0.00	0.00
+19.4	0.68	2.51	2.00	+19.5	0.10	0.00
+48.0	0.73	2.15	2.00	+47.5	0.53	0.01
				mean	**0.89**	**0.01**

hybrid with NTK/gradient-norm adaptive weighting, which rescales the data and physics terms so neither dominates by parameter-gradient magnitude [27]: on the two-parameter family at the matched 12 anchors it reaches 68.9° resonance error—still $\sim 26\times$ worse than the same anchors used as pure data (2.66°). Principled gradient balancing therefore does not rescue the hybrid. We do not test a scattered-field reformulation or causal/curriculum training, and we restrict the claim to total-field formulations on this low-dimensional, data-cheap task.

The mechanism is the known PINN failure mode [25]: for an under-resolved, high-Q field the Maxwell-residual's low-data optimum is the smooth, non-resonant field, so the residual acts as an *anti-regulariser*, pulling the fit off the correct resonant interpolation that the data alone would reach—a soft-constraint Pareto tension in which the physics and data optima do not coincide. For this resonant unit-cell response, then, the effective lever is data placement, not the physics residual—a concrete electromagnetic instance of when physics-informed regularisation does not help, complementing the general failure-mode analyses with a device-level, quantified case.

The result holds in a higher-dimensional, data-scarcer regime. A fair objection is that the ablation above is on the easy one-parameter (square-patch L) family, where data is cheapest—precisely the regime least favourable to a data-free PINN. We therefore repeat it on the full *two-parameter* (L_x, L_y) geometry family (rectangular patch, h fixed), with only 12 resonance-aware anchors over the 99-point surface—higher dimension and sparser data, where a physics prior should help most. The conclusion is unchanged and in fact sharper: the data-free PINN still fails (resonance-region error 99.4°, phase essentially flat), the physics+data hybrid at the matched 12 anchors is far worse than pure data (82.2° vs 2.66°, a $\sim 31\times$ gap), and pure resonance-aware regression on the same anchors is best. The Maxwell residual remains an anti-regulariser as the geometry dimension grows; well-placed data is the lever in both the one- and two-parameter regimes. This does not claim universality across all formulations, but it removes the "only true in the easy 1-D case" caveat.

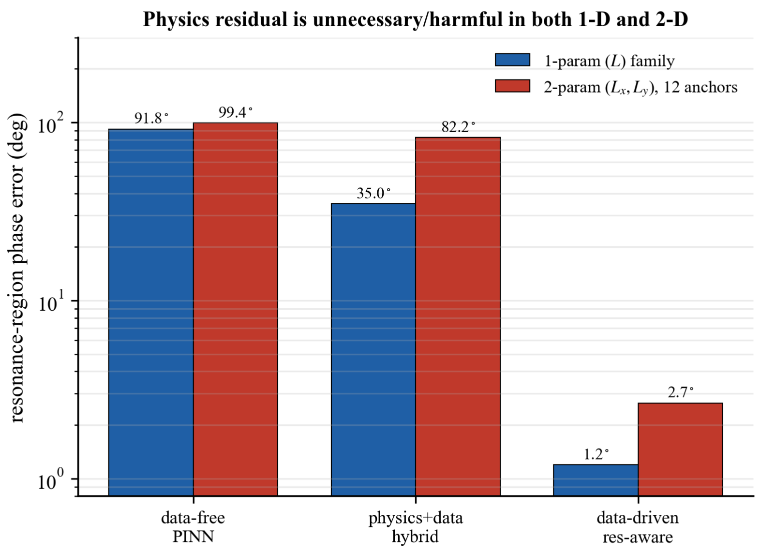

Figure 11: Ablation (resonance-region phase error, log scale) on *both* the one-parameter (L) square-patch family and the higher-dimensional, data-scarcer two-parameter (L_x, L_y) family (12 anchors). In both, a data-free PINN is flat, a physics+data hybrid is worse than pure data at matched anchors, and resonance-aware data is best—the physics residual is unnecessary and counterproductive, and the gap *widens* in 2-D ($\sim$31×). An advanced NTK/gradient-norm-balanced hybrid (not plotted) remains $\sim$26× worse than pure data, so the result is not a naive-loss-weight artefact.

13 Aperture-level demonstration: full-wave anomalous reflection

What this section delivers is the *pipeline closure*, not the grating: we test whether the surrogate→geometry→full-wave loop closes and whether the deflection *angle* is controllable by design. Aperture *efficiency* is out of scope here (and reported honestly as a limitation), because a single-resonance cell cannot deliver it; the evidence task is angle control through the differentiable model, end to end. To that end we close the loop with a full-wave *anomalous-reflection* demonstration (a coded metagrating, not a fed reflectarray). A linear reflection-phase ramp of super-period $\Lambda = Mp$ (p the cell pitch) deflects a normally-incident plane wave into the anomalous order $\theta = \arcsin(\lambda_0/\Lambda)$ by the generalized Snell law [29, 30]; choosing the super-period M at *design time* selects the deflection angle (this is geometric re-coding, not runtime/reconfigurable steering). We synthesise the ramp by inverting the surrogate's size→phase map—the same differentiable model of the preceding sections—to pick each cell size, and we close it in openEMS on a multi-super-period strip with periodic walls and a normally-incident plane wave, isolating the reflected field by a per-column two-plane decomposition and reading the diffraction orders by a spatial Fourier transform.

The full-wave anomalous order is aimed almost exactly where predicted (Table 9, Fig. 12): as M runs $8 \to 3$ its angle scans $11.3° \to 31.7°$, within $\approx 0.3°$ of generalized Snell at every point. This angle control is what a high-Q Fabry–Pérot cavity cannot provide. We report efficiency without embellishment, and note it does *not* make this a usable aperture. The anomalous order carries only $\approx$29–39% of the power, and at small M ($M = 3, 4, 5$) the specular (m=0) order actually carries *more* (52/42/44%); the anomalous lobe is correctly aimed but is not dominant there. Two causes compound: a single patch realises only 211° of reflection phase along a 1-D size cut (259° at h=1.5 mm; the full 2-D (L_x, L_y) surface spans 222°), short of the 360° a blaze needs; and—as the metasurface-efficiency literature establishes—full 360° phase coverage is *necessary but not sufficient*, because local phase-gradient (generalized-Snell) designs suffer impedance-mismatch loss that grows with deflection angle, so reaching high efficiency requires non-local/strongly-coupled elements [31]; the lossy FR4 ($\tan\delta = 0.02$, $|\Gamma|$ down to 0.68) imposes a further ceiling. The result is therefore a *proof-of-pipeline*: the surrogate-synthesised code aims the full-wave anomalous order to 0.3°, while delivering a usable-efficiency aperture is a separate (non-local element) design problem.

The per-super-period angle-and-efficiency breakdown is tabulated in Appendix C (Table 9).

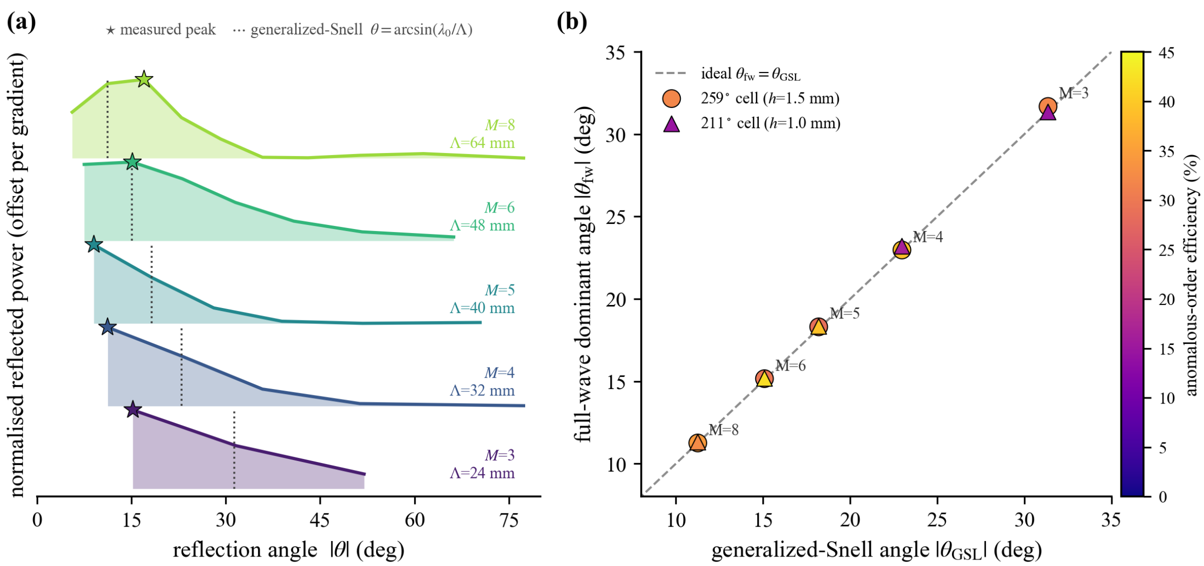

Figure 12: Full-wave anomalous reflection. Reflected power versus angle for surrogate-synthesised phase-gradient codings of decreasing super-period; the anomalous lobe scans $\approx 11° \to 32°$ and its peak tracks the generalized-Snell angle (markers). The steer angle is re-codable by the coding period.

14 Discussion

We draw out three messages. First, for sub-wavelength resonant cells the dominant cost—full-wave characterisation—is minimised not by a cleverer learner but by *where* the few samples are placed: resonance-aware active sampling cuts the resonance-region error by an order of magnitude at fixed budget and steepens the error-versus-budget law from $N^{-0.8}$ to $\approx N^{-2.3}$. This specialises general error-driven active learning for photonic surrogates [13] to a simple, resonance-targeted curvature/uncertainty criterion—demonstrated here on the microwave reflectarray cell and, changing only the oracle, shown to transfer to non-electromagnetic resonance-dominated responses (a driven-oscillator/RLC line shape and a Fano resonance, Section 8), evidence that the criterion is not specific to electromagnetics—and pairs it with a *differentiable* model so the same surrogate serves inverse design. Second, the sampling strategy and the learner are coupled: the active anchors that help the differentiable network actively hurt a global-kernel support-vector model, so "use active learning" is incomplete advice without "with a learner that tolerates clustered data." Third, the physics residual—attractive because it needs no data—is the wrong tool for this response: it fails alone and degrades the data fit when added, consistent with PINN stiffness and spectral-bias analyses [26, 28]. We are careful not to overgeneralise: this is established for a *low-dimensional, total-field* formulation of one resonant cell, a regime where data is cheap and pure regression is already strong; a scattered-field reformulation (see the outlook below) may avoid the anti-regularisation, and the PINN value proposition (no data, higher dimension) is not what this testbed probes. The conditional lesson—for this class of low-dimensional resonant responses, a total-field Maxwell residual does not help and can hurt—is what we claim.

Scope and threats to validity. We state the boundaries of every claim explicitly. (1) The study is *simulation-only* with a full-wave (not measured) baseline; no measured hardware is reported. (2) A *single* full-wave solver (openEMS) is the reference throughout, so solver-specific bias is not ruled out, though it is independently spot-checked (a square-patch cross-check agreeing to $\sim 4^\circ$ and the fine finite-difference gradient audit of Section 10). (3) The physics-residual ablation uses a *total-field* formulation only; a scattered-field reformulation is untested and may behave differently. (4) The off-electromagnetic transfer (Section 8) uses *analytic* oracles, so it demonstrates that the acquisition *criterion* transfers, *not* that it saves cost on an expensive non-EM solver; the cost-saving claims are confined to the simulated reflectarray testbed. (5) The aperture-level result is a *proof-of-pipeline* (angle correct to $\approx 0.3^\circ$; anomalous-order efficiency 29–39%), not a usable-aperture claim. (6) A noise-robustness probe shows the curvature criterion stays resonance-concentrating and keeps beating uniform sampling up to ~ 10–20% relative phase noise on the oracle, but at ~ 40% noise it ceases to concentrate on the band (anchor placement degrades toward chance) and is overtaken by uniform—so the method assumes an oracle whose curvature signal is not swamped by noise. Within these bounds, the quantified claims carry their uncertainty: every reported scaling exponent is given with a bootstrap 95% confidence interval over seeds (Tables 4, 5), and the data-efficiency dominance is stated as holding outside the per-budget ±std bands rather than as a bare point estimate.

For a *sampling-and-surrogate methodology* the full-wave reference is the correct and standard one—the method's job is to reproduce the solver from few calls, exactly as in prior active-learning surrogate work that validates against the solver [13]—and hardware measurement is the natural next milestone rather than a gap in the present claim. Beyond this, the cell is parameterised by two dimensions, and the magnitude near resonance is intrinsically less controllable than the phase because $|\Gamma|$ physically dips there. Natural extensions are higher-dimensional cells (adding, e.g., a slot or a dielectric-thickness degree of freedom, where the differentiable gradient and curvature-driven sampling should matter more), a full coded-aperture closure that chains the cell surrogate into an array-level inverse design, and a reformulated physics term acting on the scattered (rather than total) field, which may avoid the anti-regularisation identified here.

15 Conclusion

A lightweight, data-driven, *differentiable* surrogate trained on $O(10)$ full-wave anchors placed by resonance-aware active sampling models the complete two-parameter reflection-phase surface of a 24-GHz reflectarray cell, beats standard surrogates at equal sample budgets under active sampling, supplies full-wave-consistent geometry gradients, and drives a two-parameter inverse-design loop to 0.9° accuracy. A controlled ablation shows the Maxwell-residual term is unnecessary and, for this resonant response, counterproductive—the lever is data placement. The outcome is a transferable methodology—resonance-aware active learning plus a differentiable surrogate, with a quantified data-efficiency law and a device-level negative result on physics-informed regularisation—for expensive-to-sample, resonance-dominated parametric responses, here instantiated and validated end to end as a data-efficient, differentiable unit-cell design tool, with each method choice justified by direct comparison. We demonstrate it on one reflectarray cell class in simulation and claim no universality beyond that.

CRediT authorship contribution statement

Long Zhang: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Supervision, Funding acquisition. **Xuan Shi**: Software, Formal analysis, Validation. **Liang Ma**: Data curation, Visualization. **Shengjun Wu**: Methodology, Resources. **Kai You**: Validation, Resources. **Zijuan Wu**: Validation, Writing – review & editing.

Declaration of competing interest

The authors declare no competing interest.

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Ethics and consent

Not applicable.

Data availability

The openEMS full-wave reference (the 99-point 2-D and 324-point 3-D (L_x, L_y, h) grids and the metagrating closures), the trained surrogate, the active-sampling/baseline/gradient/inverse scripts, and the figure/table generators are openly available at `https://github.com/zlera20230100/reflectarray-surrogate` and archived on Zenodo (DOI: 10.5281/zenodo.20809888).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large-language-model-based assistant *solely to improve the language and readability* of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No generative-AI tool was used to generate scientific content, data, figures, or analysis, and no AI tool is listed as an author.

A Full-wave reference: openEMS configuration

The reference surfaces are computed with openEMS (FDTD) in a waveguide-simulator unit cell. Excitation: a Gaussian pulse centred at $f_0 = 24\,\text{GHz}$ with 12 GHz half-bandwidth, applied as a y-polarised plane-wave field source on a plane above the cell; fields are read in the frequency domain at f_0. Boundaries realise normal incidence with $\mathbf{E} \parallel y$: magnetic walls (PMC) $\perp x$, electric walls (PEC) $\perp y$, a PEC ground plane at $z = 0$, and an 8-cell PML above the cell. The run terminates at an energy-decay criterion of 10^{-5} (cap 6×10^4 time steps). The mesh pins explicit lines to the patch edges ($\pm L_x/2, \pm L_y/2$) and to the two readout planes, is smoothed to a maximum adjacent-cell ratio of 0.35, and near-coincident lines ($< 10^{-3}$ mm) are de-duplicated to avoid sliver cells and tiny time steps. Materials: grounded FR4 ($\varepsilon_r = 4.4$, $\tan\delta = 0.02$, conductivity $\kappa = 2\pi f_0 \varepsilon_0 \varepsilon_r \tan\delta$), thickness $h = 1\,\text{mm}$, period $p = 8\,\text{mm}$; PEC patch and ground. The reflection coefficient is extracted by sampling the mean E_y on two air-region planes $z_1 = 5\,\text{mm}$, $z_2 = 6.5\,\text{mm}$ (each $> 0.3\lambda_0$ above the patch) and solving the traveling-wave system of Eq. (1) for $\Gamma = A/B$, referenced to the ground plane. The 2-D sweep is $L_y \in \text{linspace}(2, 6, 11)$, $L_x \in \text{linspace}(2, 6, 9) = 99$ simulations; the 3-D study adds h on a $9 \times 9 \times 4 = 324$ grid; the cross-dipole and dual-resonance cells use the same simulator with the metal redefined. Exact scripts are in the released repository.

B Total-field PINN ablation: residual and boundary treatment

The ablation (Section 12) models a geometry-conditioned *scalar total field* $E_y(x, y, z; L)$ with an MLP and minimises the residual of the scalar Helmholtz equation—the y-component of the curl–curl equation under the waveguide-simulator polarisation—in the air/dielectric region,

$$\mathcal{R}(x, y, z; L) = \nabla^2 E_y + k_0^2\, \varepsilon_r(x, y, z; L)\, E_y, \tag{4}$$

where the cell geometry enters through a (smoothed) permittivity indicator $\varepsilon_r(\cdot; L)$ that encodes the patch/dielectric region as a function of the size parameters L. Boundary and excitation terms are imposed as soft penalties: $E_y = 0$ on the PEC ground plane and patch metal, the waveguide-simulator wall conditions (PMC $\perp x$, PEC $\perp y$), and an incident-field condition at the top plane reproducing the plane-wave drive; the reflection Γ is read from the network by the *same* two-plane decomposition (Eq. (1)) used for the full-wave reference, so the data loss (Eq. (2)) and the physics residual share one observable. The data-free PINN minimises Eq. (4) plus boundary/excitation terms only; the hybrid adds Eq. (2) on the anchor set; the advanced hybrid additionally rescales the data and physics terms by NTK/gradient-norm adaptive weights [27]. Collocation points are a structured grid over the air+dielectric region, de-weighted near the source plane. Loss weights, physics-weight annealing, and collocation counts were swept (Section 12); the only adjustments that improved the hybrid were those that suppressed the physics term. All hyperparameters are in the released code.

C Anomalous-reflection closure: per-super-period breakdown

Table 9 gives the full angle-and-efficiency data for the anomalous-reflection closure of Section 13. The anomalous-order angle tracks generalized Snell's law to $\approx 0.3^\circ$ at every super-period; efficiency is reported honestly: at small super-period (M=3, 4, 5) the specular (m=0) order carries *more* power than the aimed anomalous order—the angle is correct but not dominant, consistent with the single-resonance phase coverage and the non-local-element requirement discussed in Section 13.

Table 9: Full-wave anomalous reflection by a surrogate-synthesised phase-gradient metagrating (259° cell, $h = 1.5$ mm); angles as magnitudes, the order sign alternating with M.

M	Λ (mm)	GSL angle (deg)	full-wave angle (deg)	anomalous eff.	specular (m=0)
8	64	11.26	11.25	35%	10%
6	48	15.08	15.16	30%	25%
5	40	18.20	18.30	29%	44%
4	32	22.98	22.97	39%	42%
3	24	31.36	31.69	32%	52%

References

[1] G. von Trentini, "Partially reflecting sheet arrays," *IRE Trans. Antennas Propag.* 4(4):666–671, 1956.

[2] T.J. Cui, M.Q. Qi, X. Wan, J. Zhao, Q. Cheng, "Coding metamaterials, digital metamaterials and programmable metamaterials," *Light Sci. Appl.* 3:e218, 2014.

[3] A.V. Diebold, D. Pande, C. Gregg, D.R. Smith, "Reflectarray design using a discrete dipole framework," arXiv:2206.00095, 2022.

[4] W. Li, T. Qiu, J. Wang, L. Zheng, Y. Jing, Y. Jia, H. Wang, Y. Han, S. Qu, "Programmable coding metasurface reflector for reconfigurable multibeam antenna application," *IEEE Trans. Antennas Propag.* 69(1):296–301, 2021.

[5] A.H. Abdelrahman, P. Nayeri, A.Z. Elsherbeni, F. Yang, "Bandwidth improvement methods of transmitarray antennas," *IEEE Trans. Antennas Propag.* 63(7):2946–2954, 2015.

[6] A. Epstein, G.V. Eleftheriades, "Passive lossless Huygens metasurfaces for conversion of arbitrary source field to directive radiation," *IEEE Trans. Antennas Propag.* 62(11):5680–5695, 2014.

[7] L. Lu, P. Jin, G. Pang, Z. Zhang, G.E. Karniadakis, "Learning nonlinear operators via DeepONet," *Nat. Mach. Intell.* 3:218–229, 2021.

[8] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, A. Anandkumar, "Fourier neural operator for parametric partial differential equations," *ICLR*, 2021 (arXiv:2010.08895).

[9] Y. Augenstein, T. Repän, C. Rockstuhl, "Neural operator-based surrogate solver for free-form electromagnetic inverse design," *ACS Photonics* 10(5):1547–1557, 2023.

[10] J. Jiang, M. Chen, J.A. Fan, "Deep neural networks for the evaluation and design of photonic devices," *Nat. Rev. Mater.* 6:679–700, 2021.

[11] S. Koziel, A. Pietrenko-Dabrowska, L. Leifsson, "Antenna optimization using machine learning with reduced-dimensionality surrogates," *Sci. Rep.* 14:21567, 2024.

[12] N. Sarker, P. Podder, M.R.H. Mondal, S.S. Shafin, J. Kamruzzaman, "Applications of machine learning and deep learning in antenna design, optimization, and selection: a review," *IEEE Access* 11:103890–103915, 2023.

[13] R. Pestourie, Y. Mroueh, T.V. Nguyen, P. Das, S.G. Johnson, "Active learning of deep surrogates for PDEs: application to metasurface design," *npj Comput. Mater.* 6:164, 2020 (arXiv:2008.12649).

[14] B. Settles, *Active Learning*, Synthesis Lectures on Artificial Intelligence and Machine Learning, Morgan & Claypool, 2012.

[15] T. Lookman, P.V. Balachandran, D. Xue, R. Yuan, "Active learning in materials science with emphasis on adaptive sampling using uncertainties for targeted design," *npj Comput. Mater.* 5:21, 2019.

[16] K. Crombecq, D. Gorissen, D. Deschrijver, T. Dhaene, "A novel hybrid sequential design strategy for global surrogate modeling of computer experiments," *SIAM J. Sci. Comput.* 33(4):1948–1974, 2011.

[17] R.B. Gramacy, *Surrogates: Gaussian Process Modeling, Design, and Optimization for the Applied Sciences*, Chapman & Hall/CRC, 2020.

[18] O. Sener, S. Savarese, "Active learning for convolutional neural networks: a core-set approach," in *Int. Conf. on Learning Representations (ICLR)*, 2018.

[19] D. Liu, Y. Tan, E. Khoram, Z. Yu, "Training deep neural networks for the inverse design of nanophotonic structures," *ACS Photonics* 5(4):1365–1369, 2018.

[20] S. Molesky, Z. Lin, A.Y. Piggott, W. Jin, J. Vučković, A.W. Rodriguez, "Inverse design in nanophotonics," *Nat. Photonics* 12:659–670, 2018.

[21] M. Raissi, P. Perdikaris, G.E. Karniadakis, "Physics-informed neural networks," *J. Comput. Phys.* 378:686–707, 2019.

[22] G.E. Karniadakis, I.G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, L. Yang, "Physics-informed machine learning," *Nat. Rev. Phys.* 3:422–440, 2021.

[23] A. Beltrán-Pulido, I. Bilionis, D. Aliprantis, "Physics-informed neural networks for solving parametric magnetostatic problems," *IEEE Trans. Energy Convers.* 37(4):2678–2689, 2022.

[24] M. Nohra, S. Dufour, "Physics-informed neural networks for the numerical modeling of steady-state and transient electromagnetic problems with discontinuous media," arXiv:2406.04380, 2024.

[25] A.S. Krishnapriyan, A. Gholami, S. Zhe, R.M. Kirby, M.W. Mahoney, "Characterizing possible failure modes in physics-informed neural networks," *NeurIPS*, 2021.

[26] S. Wang, X. Yu, P. Perdikaris, "When and why PINNs fail to train: a neural tangent kernel perspective," *J. Comput. Phys.* 449:110768, 2022.

[27] S. Wang, Y. Teng, P. Perdikaris, "Understanding and mitigating gradient flow pathologies in physics-informed neural networks," *SIAM J. Sci. Comput.* 43(5):A3055–A3081, 2021.

[28] M. Tancik et al., "Fourier features let networks learn high frequency functions in low dimensional domains," *NeurIPS* 33:7537–7547, 2020.

[29] N. Yu, P. Genevet, M.A. Kats, F. Aieta, J.-P. Tetienne, F. Capasso, Z. Gaburro, "Light propagation with phase discontinuities: generalized laws of reflection and refraction," *Science* 334(6054):333–337, 2011.

[30] S. Sun, Q. He, S. Xiao, Q. Xu, X. Li, L. Zhou, "Gradient-index meta-surfaces as a bridge linking propagating waves and surface waves," *Nat. Mater.* 11(5):426–431, 2012.

[31] A. Díaz-Rubio, V.S. Asadchy, A. Elsakka, S.A. Tretyakov, "From the generalized reflection law to the realization of perfect anomalous reflectors," *Sci. Adv.* 3(8):e1602714, 2017.

[32] D.M. Pozar, S.D. Targonski, H.D. Syrigos, "Design of millimeter wave microstrip reflectarrays," *IEEE Trans. Antennas Propag.* 45(2):287–296, 1997.

[33] P. Robustillo, J. Zapata, J.A. Encinar, J. Rubio, "ANN characterization of multi-layer reflectarray elements for contoured-beam space antennas in the Ku-band," *IEEE Trans. Antennas Propag.* 60(7):3205–3214, 2012.

[34] D.R. Prado, J.A. López-Fernández, G. Barquero, M. Arrebola, F. Las-Heras, "Fast and accurate modeling of dual-polarized reflectarray unit cells using support vector machines," *IEEE Trans. Antennas Propag.* 66(3):1258–1270, 2018.